

Hr_Analytics Business Report

A.) Workforce Overview

-1. What does our current workforce look like — headcount, composition, and structure?

Answer;

The organization has 311 total employees, 207 of whom are currently active, spread across 6 departments. The workforce is 56.6% female and 43.4% male — a 13.2 percentage-point gap. Racial composition spans 6 categories, with White employees the largest group; representation narrows sharply toward the smaller categories, including a single Hispanic employee.

-2. How is headcount distributed across departments and positions?

Answer;

Production is the organization's largest department by a wide margin, with 209 employees — 67.2% of total headcount — functioning as the organization's primary labour force. Within Production, Production Technician I alone accounts for 65.6% of the department's staff.

Admin Offices holds the smallest headcount among standard operating departments at 9 employees; Executive Office holds the smallest overall at 1 — its sole occupant is the President & CEO, reporting directly to the Board of Directors, which is a normal structure at the top of any org chart rather than a data gap.

-3. How has hiring activity changed over time?

Answer;

Hiring grew gradually from 2006–2010, then spiked sharply to 83 new hires in 2011 — the organization's single largest hiring cohort. Activity fluctuated downward through 2012–2017, tapering to just 1 hire in 2018.

Notably, this 2011 cohort later explains a large share of the 2015–2016 departure spike (see Turnover section) — roughly half of all departures in those two years came from employees hired in 2011, reaching their 4–5 year tenure mark at that point.

-4. Are there measurable pay differences by gender or race?

Answer;

At the company level, men earn 4.2% more than women on average — a gap worth monitoring but within a range often attributable to role and tenure mix rather than direct pay inequity.

Breaking this down by department reveals a more nuanced picture: men earn more in Admin, IT/IS, and Production (averaging a 3.1% gap across the three), while women earn more in Sales (8.6%) and Software Engineering (4.5%) — meaning the company-wide figure obscures two departments where the gap runs in the opposite direction.

Average tenure shows no meaningful gender difference (4.82 vs. 4.83 years), ruling out tenure as an explanation for the pay gaps observed.

Race-based salary comparisons are limited by extreme sample imbalance — one category (Hispanic) reflects a single employee's salary (Production Manager Brannon Miller) and cannot be generalized.

B.) Turnover & Retention

-1. What is our overall employee attrition rate, and how much of it is voluntary vs. involuntary?

Answer;

Overall attrition rate = 33.4%

Voluntary = 28.3%

Involuntary = 4.5%

Unclassified = 0.6% (Due to data quality issues)

-2. What are the primary reasons employees leave?

Answer;

Career Growth (29 departures) and Dissatisfaction (22) are the two leading reasons employees leave — together accounting for roughly half of all voluntary departures.

This pattern points toward questions worth raising internally: are there sufficient internal advancement paths, and is the organization's training/development investment keeping pace with employee ambition? These are hypotheses the data can point toward, not conclusions it can confirm on its own.

-3. Is turnover concentrated in the first 1–3 years of tenure?

Answer;

Yes. < 1 Year Headcount = 4, terminated = 4. 1-3 yrs; Headcount = 45, Terminated = 35.

-4. Are turnover rates significantly different across departments and managers?

Answer;

Yes, attrition varies significantly by both. By department, Production (39.7%) and Software Engineering (36.4%) show the highest attrition.

By manager, the pattern is sharper still: Amy Dunn and Webster Butler each show 61.9% attrition across meaningfully sized teams (21 direct reports each) — a genuine, statistically credible hotspot.

One manager shows 100% attrition, but on a team of exactly 1 — not a pattern, just one individual's departure, and shouldn't be read as a management signal.

-5. How has the volume of departures changed year over year?

Answer;

2015 and 2016 recorded the highest departure volumes (23 and 22 respectively). This aligns closely with the 2011 hiring spike — roughly half of departures in both peak years came from that single cohort reaching its 4–5 year tenure mark, suggesting the spike is partly a function of hiring-cohort size rather than a standalone retention failure specific to those two years.

C.) Engagement, Performance & Recruitment

-1. Do engagement, satisfaction, absenteeism, or lateness differ between employees who stayed and those who left?

Answer;

Self-reported attitude metrics show no measurable difference between employees who stayed and those who left — engagement and satisfaction scores are effectively identical across both groups.

Behavioural metrics tell a different story: terminated employees show meaningfully higher absenteeism. This suggests attitude surveys alone are weak predictors of attrition risk here, while attendance patterns may be a more useful early signal worth monitoring.

-2. Which recruitment sources produce employees who stay longer, perform better, and report higher engagement?

Answer;

Website and Employee Referral show the strongest combined profile — both post the lowest attrition rates among any adequately-sized source (roughly 14% and 16% respectively) alongside solid engagement scores.

Google Search, despite comparable engagement scores to the stronger sources, shows by far the weakest retention of any channel with meaningful volume (61.2% attrition, n=49).

Recommendation; doubling down on Website and Employee Referral specifically, and flag Google Search for review rather than expansion.

-3. Is there a relationship between performance rating, salary, and retention?

Answer;

Employees under the exceeds performance band has the highest average salary of 81k (for active employees.)

While fully meets band has the least salary average of 59K (active) 56K (Term).

The other 2 bands are between 64k-74K (both active and term.).

Observation/question; Why does fully meet band have the lowest avg salary? - salary in this organization tracks job role and level far more than it tracks performance rating.

78% of the workforce is currently coded as PIP — this warrants validation. Does this reflect the field being repurposed or mislabelled in the source HR system over time, or a genuine, organization-wide performance concern?

Recommendations;

1. Investigate Production and Software Engineering retention — the highest department-level attrition in the two largest operational/technical teams.
2. Review management practices under the two managers with 61.9% attrition on substantial teams — the clearest credible hotspot in the data.
3. Shift recruiting investment toward Website and Employee Referral channels; audit the Google Search hiring pipeline given its disproportionate attrition.
4. Strengthen first 1–3 year onboarding and retention support, where departure risk is highest across the organization.
5. Validate Performance Score data with HR before further use, given the unusually high proportion of employees coded "PIP."